ALGEBRA 1 · POLYNOMIALS

Area Model Multiplier

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Area Model Multiplier** on Mr. K's Classroom: Algebra 1 → Polynomials → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Is $(x + 3)^2$ equal to $x^2 + 9$? Explain.

   No. The rectangle also has two 3x pieces: $(x + 3)^2 = x^2 + 6x + 9$.

2. How many different kinds of pieces does the rectangle for $(x + 2)(x + 5)$ have?

   Three: x^2, x-pieces (5x and 2x) and number pieces (10).

2 Explore use the tool and record what you see

Build each product

Record each part of the rectangle, then combine.

Product	x^2 part	x parts	Number part	Simplified
$(x + 2)(x + 3)$	x^2	3x and 2x	6	$x^2 + 5x + 6$
$(x + 4)(x - 1)$	x^2	−x and 4x	−4	$x^2 + 3x - 4$
$(x + 3)^2$	x^2	3x and 3x	9	$x^2 + 6x + 9$
$(x + 5)(x - 5)$	x^2	−5x and 5x	−25	$x^2 - 25$
$(2x + 1)(x + 3)$	$2x^2$	6x and x	3	$2x^2 + 7x + 3$

3 Write the rule fill in each blank from what you observed

1. The middle term comes from **adding** the two x parts.
2. The constant term is the **product** of the two constants.
3. $(x + a)^2 = x^2 +$ **2ax** $+ a^2$, not $x^2 + a^2$.
4. $(x + a)(x - a) =$ **$x^2 - a^2$**, because the two x parts cancel.

4 Check no tool: show your work

1. Multiply $(x + 6)(x + 2)$.

   $x^2 + 8x + 12$

2. Multiply $(x - 3)(x - 7)$.

   $x^2 - 10x + 21$

3. Multiply $(x - 4)^2$.

   $x^2 - 8x + 16$

4. Multiply $(3x + 2)(x - 1)$.

   $3x^2 - x - 2$

5 Reflect

Draw or describe the rectangle for $(x + 1)(x + 4)$ and explain where each term comes from.

$x \cdot x = x^2$, $x \cdot 4 = 4x$, $1 \cdot x = x$ and $1 \cdot 4 = 4$, so the product is $x^2 + 5x + 4$.